

Varun Goud Karupothula

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SUMMARY

Final-year Computer Science student combining a strong technical foundation with hands-on experience in software development, data analytics, and AI integration. Proven track record of building production-style applications using React and FastAPI, optimizing data pipelines for sub-millisecond retrieval, and leveraging advanced AI models to solve complex user problems. Certified OCI Generative AI Professional adept at translating data insights into scalable product features.

SKILLS

Programming & Frameworks	: Python, Java, JavaScript, FastAPI, React, HTML, CSS
Data & AI Engineering	: Pandas, NumPy, Scikit-learn, TensorFlow, FAISS, PostgreSQL, MySQL, MongoDB, Supabase
Tools & Methodologies	: Git, GitHub, VS Code, Vite, Vercel, Data Analytics, Agile Concepts
Core CS Foundations	: Data Structures & Algorithms, DBMS, Object-Oriented Programming (OOP), Operating Systems, Computer Networks
Core Competencies	: Analytical Thinking, Problem Solving, Cross-Functional Communication, Teamwork

PROJECTS

Studyys – AI-Powered Study Assistant

- Developed a **document-aware study assistant** with custom exam compilers, active recall simulators, and a peer-to-peer library.
- Integrated **Google Gemini 1.5 Flash** for a contextual chat engine retrieving textbook definitions, cutting answer latency by **25%**.
- Implemented client-side extraction using **PDF.js** and **Web Workers**, parsing documents in **under 5 seconds** with no server load.
- Designed a **PostgreSQL** schema on **Supabase** with 15 normalized tables, custom triggers, and **Row Level Security (RLS)**.

[Github](#) [Live](#)

Semantic Cache Search Engine – Cluster-Aware Document Retrieval

- Built a **cluster-aware semantic search engine** with GMM-based cache routing, a FAISS vector index, and a React dashboard.
- Encoded queries into **384-dim embeddings** via **SentenceTransformers**, clustering with **GMM** to achieve a **35–45% cache hit rate**.
- Built a CPU-optimized **FAISS (IndexFlatIP)** index over **20,000+ documents** for sub-millisecond cosine-similarity search.
- Deployed via **React/Vite**; cache hits above **0.85 similarity** cut avg. API response time by **80%**.

[Github](#) [Live](#)

Heart Disease Detection using CNN and LSTM

- Built a **5-phase deep learning pipeline** for **ECG heartbeat classification** on the **MIT-BIH Arrhythmia Database**, mapped to **AAMI categories**.
- Designed a **Hybrid 1D-CNN + LSTM** architecture — CNNs extract waveform features (P-waves, QRS complexes); LSTM captures temporal patterns.
- Applied **Butterworth bandpass filtering** and class-imbalance handling; built a **CV engine (OpenCV)** for image-based arrhythmia classification.
- Achieved **98.7% accuracy** and **97.4% F1-score**, outperforming CNN (94.2%) and LSTM (91.8%) baselines, matching the **Yildirim (2020)** benchmark.

[Github](#)

EDUCATION

Vellore Institute of Technology Amaravathi B.Tech in Computer Science & Engineering	Sep 2023 - Jun 2027 CGPA : 8.25
Sri Chaitanya Junior College Hyderabad Intermediate (Class XII)	Jun 2020 - Jul 2022 Percentage : 95.1
Dr. K.K.R's Gowtham School Hyderabad Secondary School Certificate (Class X)	Jul 2019 - Apr 2020 CGPA : 10

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional 	2025
ORACLE	
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate 	2025
ORACLE	